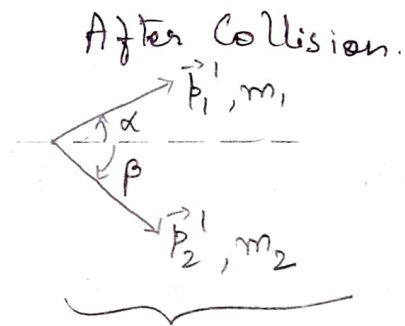
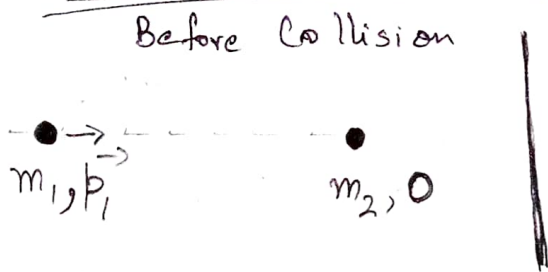
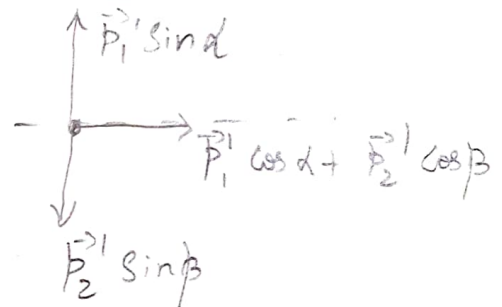


Collision Kinematics



Using Conservation of momentum before & after Collision.

Resolving the momentum.



$$\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$$

$$\therefore \vec{p}_1 = \vec{p}_1' + \vec{p}_2' \quad \left. \begin{array}{l} \text{Resolving this equation} \\ \text{for x \& y direction} \end{array} \right\}$$

Now conservation of momentum in x direction

$$\vec{p}_1 = p_1' \cos \alpha + p_2' \cos \beta$$

Conservation of momentum in y direction

$$0 = p_1' \sin \alpha - p_2' \sin \beta$$

Now conservation of Energy.

$$\frac{p_1^2}{2m_1} = \frac{(p_1')^2}{2m_1} + \frac{(p_2')^2}{2m_2}$$

Using $KE = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m} = \frac{p^2}{2m}$

(2)

In the three equations above

Known is p_1
Unknown are p_1' , p_2' , α & β } $\therefore 3$ eqns
4 unknowns

$\therefore$ We Cannot Solve these equation. To solve such system, assume " β " is known and use the fact that particle/mass 2 cannot move backward $\therefore \beta \leq \frac{\pi}{2}$.

$\therefore$ we can now find p_1' , p_2' & α in terms of p_1 & β .

$\rightarrow$ Now let $x = \frac{p_1'}{p_1}$ $y = \frac{p_2'}{p_1}$ & $r = \frac{m_1}{m_2}$

$\therefore$ from conservation of momentum in Y dirn } $p_1' \sin \alpha - p_2' \sin \beta = 0$

dividing both sides by p_1 } $\frac{p_1'}{p_1} \sin \alpha = \frac{p_2'}{p_1} \sin \beta$

$$x \sin \alpha = y \sin \beta \quad \text{--- (1)}$$

from conservation of momentum in X dirn } $p_1 = p_1' \cos \alpha + p_2' \cos \beta$

$$1 = \frac{p_1'}{p_1} \cos \alpha + \frac{p_2'}{p_1} \cos \beta$$

$$\therefore 1 = x \cos \alpha + y \cos \beta$$

$$\therefore x \cos \alpha = 1 - y \cos \beta \quad \text{--- (2)}$$

from energy conservation equ.

$$\frac{p_1^2}{2m_1} = \frac{p_1'^2}{2m_1} + \frac{p_2'^2}{2m_2}$$

$$1 = \left(\frac{p_1'}{p_1} \right)^2 + \left(\frac{p_2'}{p_1} \right)^2 \frac{m_1}{m_2}$$

$$x^2 + y^2 = 1 \quad \text{--- (3)}$$

Square and add equations (1) & (2)

$$x^2 \sin^2 \alpha + x^2 \cos^2 \alpha = y^2 \sin^2 \beta + (1 - y \cos \beta)^2$$

$$x^2 (\sin^2 \alpha + \cos^2 \alpha) = y^2 \sin^2 \beta + 1 + y^2 \cos^2 \beta - 2y \cos \beta$$

$$x^2 = y^2 + 1 - 2y \cos \beta$$

from eqn (3) we have $x^2 = 1 - ry^2$

$$\therefore 1 - ry^2 = y^2 + 1 - 2y \cos \beta$$

$$2y \cos \beta = y^2 (1 + r)$$

$$\therefore \boxed{y = \frac{2 \cos \beta}{1+r}}$$

Now use this is eqn (3)

$$x^2 = 1 - ry^2 = 1 - r \left(\frac{2 \cos \beta}{1+r} \right)^2$$

$$x^2 = \frac{1 + 2r - 4r \cos^2 \beta + r^2}{(1+r)^2}$$

$$x = \frac{\sqrt{1 - 2r \cos 2\beta + r^2}}{1+r} \quad \left\{ \begin{array}{l} \text{using} \\ \downarrow \\ 2 \cos^2 \beta - 1 = \cos 2\beta \end{array} \right.$$

Now divide eqn (1) by (2)

$$\frac{x \sin \alpha}{x \cos \alpha} = \frac{y \sin \beta}{1 - y \cos \beta}$$

$$\tan \alpha = \frac{y \sin \beta}{1 - y \cos \beta} = \frac{\left(\frac{2 \cos \beta}{1+r} \right) \sin \beta}{1 - \left(\frac{2 \cos \beta}{1+r} \right) \cos \beta}$$

$$\tan \alpha = \frac{2 \sin \beta \cos \beta}{1+r - 2 \cos^2 \beta} = \frac{\sin 2\beta}{r - \cos 2\beta}$$

∴ we have the following 3 results (5)

$$y = \frac{2 \cos \beta}{1+r}$$

$$x = \frac{\sqrt{1 - 2r \cos 2\beta + r^2}}{1+r}$$

$$\tan d = \frac{\sin 2\beta}{r - \cos 2\beta}$$

using $\frac{m_1}{m_2} = r$ $x = \frac{p_1'}{p_1}$ $y = \frac{p_2'}{p_1}$

These can be re-written as

$$\tan d = \frac{m_2 \sin 2\beta}{m_1 - m_2 \cos 2\beta}$$

$$p_1' = \frac{p_1 \sqrt{m_2^2 - 2m_1 m_2 \cos 2\beta + m_1^2}}{(m_1 + m_2)}$$

$$p_2' = \frac{2m_2 p_1 \cos \beta}{m_1 + m_2}$$

Special Cases

Case 1 : $\beta = 0$

If $\beta = 0$ Then
$$\frac{\sin 2\beta}{\gamma - \cos 2\beta} = \frac{0}{\gamma - 1}$$

$\tan \alpha = 0$
 $\Rightarrow \alpha = 0$



Before Collision

After Collision
 (straight line motion)

$$\kappa = \frac{\sqrt{\gamma^2 - 2\gamma \cos 2\beta + 1}}{1 + \gamma} = \frac{\sqrt{\gamma^2 - 2\gamma + 1}}{1 + \gamma}$$

$$\kappa = \frac{\sqrt{(\gamma - 1)^2}}{\gamma + 1}$$

$$\kappa = \frac{\gamma - 1}{\gamma + 1}$$

$$\kappa = \frac{m_1 - m_2}{m_1 + m_2}$$

$$\therefore \frac{p_2'}{p_1} = \frac{m_1 - m_2}{m_1 + m_2} \quad \text{or} \quad \vec{p}_1' = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{p}_1$$

$$y = \frac{2 \cos \beta}{1+r} = \frac{2 \cos 0}{1+r}$$

$$\therefore \frac{p_2'}{p_1} = \frac{2}{1 + \frac{m_1}{m_2}}$$

$$\underline{\underline{p_2' = \left(\frac{2m_2}{m_1 + m_2} \right) p_1}}$$

Case 2 : $r=1 \Rightarrow m_1 = m_2$ (Equal mass case)

$$x = \frac{\sqrt{1 - 2r \cos 2\beta + r^2}}{1+r}$$

$$\Rightarrow x = \frac{\sqrt{1 - 2 \cos 2\beta + 1}}{2} = \frac{\sqrt{2(1 - \cos 2\beta)}}{2}$$

$$x = \frac{\sqrt{4 \sin^2 \beta}}{2} \quad \left(\begin{array}{l} 1 - \cos 2\beta \\ = 2 \sin^2 \beta \end{array} \right)$$

$$\underline{\underline{x = \sin \beta}}$$

$$\Rightarrow \frac{p_1'}{p_1} = \sin \beta$$

$$\underline{\underline{p_1' = p_1 \sin \beta}}$$

$$y = \frac{2 \cos \beta}{1+r} = \frac{2 \cos \beta}{1+1} = \cos \beta$$

$$\Rightarrow \frac{p_2'}{p_1} = \cos \beta$$

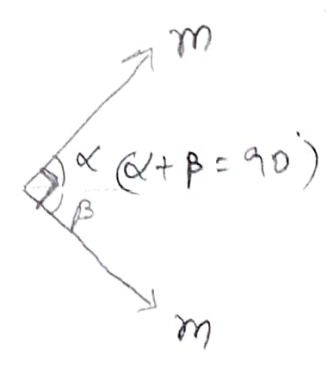
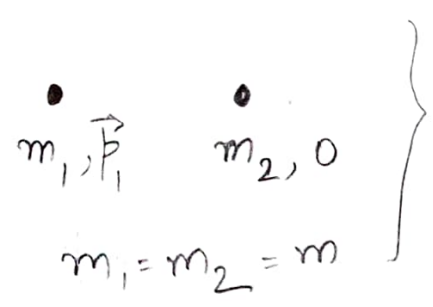
$$\therefore p_2' = p_1 \cos \beta$$

$$\tan \alpha = \frac{\sin 2\beta}{1 - \cos 2\beta} = \frac{\sin 2\beta}{1 - \cos 2\beta} = \frac{2 \sin \beta \cos \beta}{2 \sin^2 \beta}$$

$$\tan \alpha = \frac{\cos \beta}{\sin \beta} = \cot \beta$$

$$\Rightarrow \alpha = \frac{\pi}{2} - \beta$$

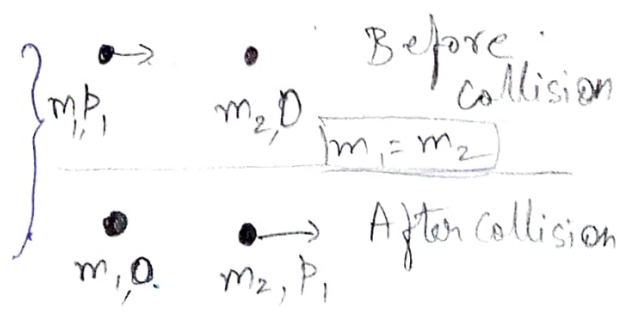
$$\therefore \alpha + \beta = \frac{\pi}{2}$$



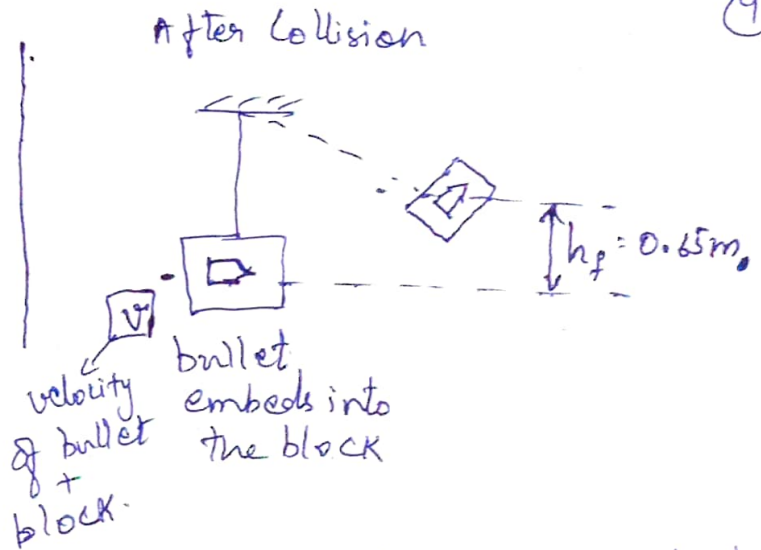
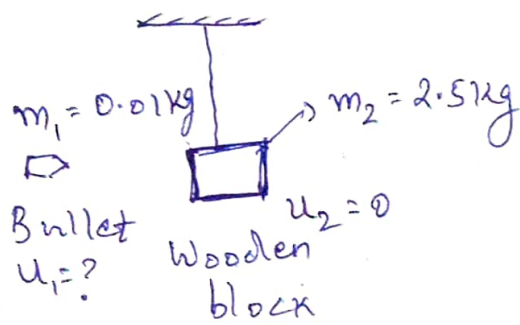
Case 3: $\alpha = \beta = 0$ & $\frac{m_1}{m_2} = 1 \Rightarrow m_1 = m_2$

Then.
Momentum Exchange

$$\left. \begin{aligned} p_1' &= p_1 \sin 0 = 0 \\ p_2' &= p_1 \cos 0 = p_1 \end{aligned} \right\}$$



1)



Given that a bullet moving towards a suspended wooden block as shown in figure (left). What is the initial velocity u_1 of the bullet if the bullet gets embedded into block & this system rises to a height $h_f = 0.65 \text{ m}$ as shown in figure on right.

Sol: Conservation of momentum for the system.

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$\underbrace{u_2}_{0 \text{ (given)}}$

$$\therefore m_1 u_1 = (m_1 + m_2) v$$

For the System after collision.

use Conservation of Energy.

$$\frac{1}{2} (m_1 + m_2) v^2 + \underbrace{mg \cdot 0}_{\text{initial PE} = 0} = \underbrace{\frac{1}{2} (m_1 + m_2) \cdot 0^2}_{\text{final velocity is zero}} + mgh_f$$

$$\therefore \frac{1}{2} (m_1 + m_2) v^2 = (m_1 + m_2) g h_f$$

$$v = \sqrt{2gh_f} = \sqrt{2 \times 9.81 \times 0.5} = \underline{\underline{3.57 \text{ m/s}}}$$

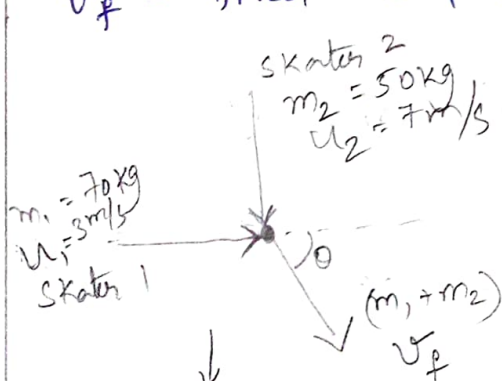
Substitute this in the momentum conservation

$$m_1 u_1 = (m_1 + m_2) v$$

$$u_1 = \frac{m_1 + m_2}{m_1} v = \left(\frac{0.01 + 2.50}{0.01} \right) 3.57$$

$$\text{The velocity of bullet is} = \underline{\underline{896.07 \text{ m/s}}}$$

- 2) A 70 kg skater is travelling east at 3 m/s speed. Another skater (50 kg) is moving south at 7 m/s. They collide and hold on to each other after the collision. They start moving at angle θ south of east with velocity v_f . find θ & v_f . Neglect friction.



Sketch of skaters before & after collision.

$$\left. \begin{aligned} m_1 u_1 &= (m_1 + m_2) v_f \cos \theta \\ m_2 u_2 &= (m_1 + m_2) v_f \sin \theta \end{aligned} \right\} \begin{array}{l} \text{Conservation of} \\ \text{Momentum} \\ \text{in x dirn.} \end{array}$$
$$\left. \begin{aligned} m_2 u_2 &= (m_1 + m_2) v_f \sin \theta \end{aligned} \right\} \begin{array}{l} \text{Conservation of} \\ \text{Momentum} \\ \text{in y dirn} \end{array}$$

divide second eqn with first

$$\tan \theta = \frac{m_2 u_2}{m_1 u_1} = \frac{50 \times 7}{70 \times 3} = 1.67$$

$$\Rightarrow \theta \approx 59^\circ$$

$$v_f = \frac{m_1 u_1}{(m_1 + m_2) \cos \theta} = \frac{70 \times 3}{(70 + 50) \cos 59^\circ} \approx \underline{\underline{3.39 \text{ m/s}}}$$